

CATS: Constraint-Aware Topology Scheduling Mechanism for Efficient LLM Training in O/E Hybrid Switching Networks

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Abstract We propose *Constraint-Aware Topology Scheduling (CATS)* mechanism to overcome the indiscriminate scheduling exclusively based on traffic volume in O/E hybrid networks. CATS strategically offloads the constraint-induced traffic to the contention-free, zero-jitter optical paths, accelerating LLM training average time by 0.49%–9.16% and boosting TFLOPS by 0.61%–9.12%. ©2026 The Author(s)

Introduction

The continuous scaling of Large Language Models (LLMs) imposes stringent communication demands on distributed training, which generates highly heterogeneous traffic [1,2]. Specifically, Data Parallelism (DP) drives bandwidth-sensitive parameter synchronization traffic, whereas Pipeline Parallelism (PP) produces sequential, latency-sensitive traffic. Traditional Electrical Packet Switches (EPS) struggle to accommodate these distinct characteristics as scaling up. Massive DP traffic inevitably induces incast congestion, triggering Explicit Congestion Notification (ECN) induced rate reductions that severely degrade effective bandwidth [3]. Conversely, the inherent store-and-forward architecture of EPS introduces queuing jitter, disrupting latency-sensitive PP traffic and prolonging GPU idle times.

To address these limitations, Optical Circuit Switching (OCS) has emerged as a promising paradigm, offering contention-free, high-bandwidth, and deterministic low-latency optical paths[4]. Recent advancements have confirmed its critical role in data centers [5,6]. The combination of OCS and EPS synergizes the deterministic, high-bandwidth capacity of the OCS with the statistical multiplexing and routing flexibility of the EPS [7]. However, existing Optical and Electrical (O/E) hybrid networks mechanism schedules optical paths exclusively based on traffic volume, allocating dedicated optical paths to massive traffic to exploit the high-bandwidth capacity of OCS [8]. The volume-centric mechanism overlooks the synchronous execution model of LLM training. By indiscriminately prioritizing massive DP traffic, the mechanism inherently neglects low-volume yet highly latency-sensitive PP traffic. This ignores the straggler effect inherent in synchronous LLM training, where even low-volume traffic can hinder the entire training efficiency.

To resolve this, we propose the Constraint-Aware Topology Scheduling (CATS) mechanism for O/E hybrid networks. Instead of indiscriminately scheduling dedicated optical paths based on traffic volume, CATS identifies the prevailing network performance constraint, which is defined as either a bandwidth-constraint by DP or a latency-constraint by PP. By reconfiguring the topology to allocate optical paths specifically for constraint-inducing traffic, CATS leverages the contention-free and jitter-zero properties of OCS to overcome the performance bottlenecks. Driven by an iteration time-cost model, CATS determines the topology prior to training. The real-world experiments demonstrate that this CATS mechanism effectively improve the training efficiency by accelerating average iteration time from 0.49% to 9.16% and ascending TFLOPS from 0.61% to 9.12%.

The principle of CATS mechanism

We propose the CATS mechanism designed for the O/E hybrid networks as illustrated in Fig.1(a). LLM training exhibits dynamic performance constraints. We abstract these constraints into two dominant regimes.

First, in the bandwidth-constraint regime, frequent and massive DP synchronizations severely degrade the effective bandwidth. Within standard EPS, 5-tuple hash mapping inevitably causes hash polarization and persistent incast on specific ports. This triggers excessive ECN marking, forcing transport protocols to aggressively reduce effective bandwidth. To mitigate this, we offload bandwidth-sensitive DP traffic to an OCS. By scheduling deterministic, contention-free optical paths, the OCS fundamentally eradicates incast contention at the hardware level as shown in Fig.1(b).

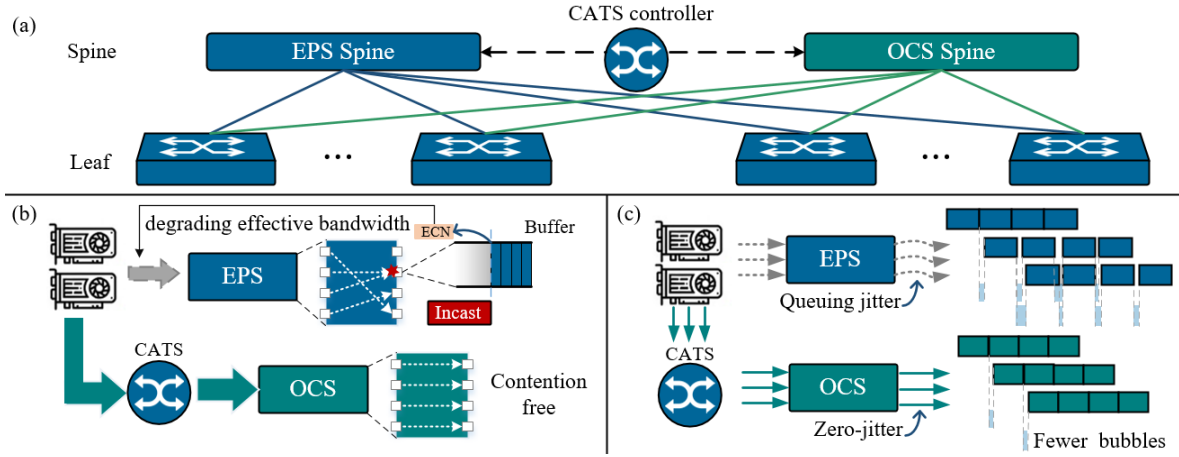


Fig. 1: (a) O/E hybrid network. (b) OCS settles bandwidth-constraints. (c) OCS settles latency-constraints.

Second, in the latency-constraint regime. Once the EPS possesses sufficient residual bandwidth to absorb DP traffic without triggering ECN, offloading DP traffic to the OCS offers no performance gains. Therefore, overall performance is dictated by the network jitter. Although PP generates low-volume traffic, it is sensitive to any microsecond-level delays, meaning that reducing latency is the efficient way to improve training efficiency. Stochastic EPS queuing variance introduces unpredictable delays that prolong downstream GPU idle times, manifesting as expanded pipeline bubbles. Strategically offloading latency-sensitive PP traffic to the zero-jitter optical domain perfectly compresses bubbles as shown in Fig.1(c).

Although OCS is effective for both bandwidth- and latency-sensitive traffic, we do not use an all-optical spine layer as the circuit switching feature lacks flexibility. This hinders scalability as packet switching shares the bandwidth source.

The procedure of CATS mechanism

The CATS mechanism schedules network topology by strategically offloading constraint-inducing traffic to the optical paths. Initially, CATS extracts the distributed training parameters pre-defined for a specific workload. Based on these extracted profiles, CATS predicts the dominant network performance constraints prior to training. By leveraging the contention-free and zero-queuing characteristics of the OCS, CATS redirects the specific traffic onto dedicated optical paths, thereby improving the training efficiency.

To support predictive scheduling, CATS employs an analytical iteration time-cost model based on a standard 1F1B pipeline schedule. Rather than simulating exhaustive execution times, the model isolates different network constraints by quantifying the differential time-cost (ΔT) between two topological configurations: offloading PP to the OCS versus offloading DP to the OCS. We define ΔT_{DP} as the macroscopic time penalty incurred by exposing massive DP All-Reduce

payloads to EPS Incast congestion, and Δt_{pp} as the microscopic latency induced per pipeline inter-stage transfer due to EPS queuing jitter. The iteration time difference cleanly reduces to the physical penalties:

$$\Delta T = \Delta T_{DP} - (p - 1 + n) \times \Delta t_{pp} \quad (1)$$

Here, p designates the pipeline stages, and n is the total micro-batch count, explicitly defined by the ratio of Global Batch Size (GBS) to the product of DP degree (dp) and Micro-Batch Size (MBS), yielding $n = \frac{GBS}{dp \cdot MBS}$ in Eq. (2). CATS mandates routing DP traffic over the OCS if and only if $\Delta T > 0$, yielding the core decision inequality:

$$\Delta T_{DP} > (p - 1 + \frac{GBS}{dp \cdot MBS}) \times \Delta t_{pp} \quad (2)$$

Extracting precise communication penalties a priori from active training is fundamentally intractable. Consequently, CATS parameterizes these constants using RDMA microbenchmarks. The microscopic PP penalty (Δt_{pp}) is captured via small-message latency differentials, isolating EPS queuing jitter. Conversely, the macroscopic DP penalty (ΔT_{DP}) utilizes large-message benchmarks to quantify EPS congestion-induced bandwidth degradation. Evaluating the Eq. (2) with these physical proxies deterministically triggers the optimal topology configuration.

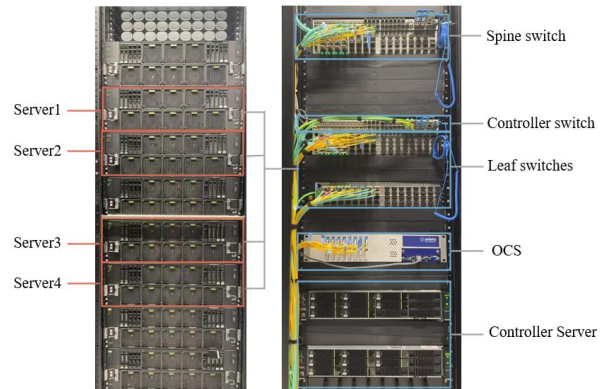


Fig. 2: Real-world O/E hybrid testbed of CATS.

Experimental evaluation and discussion

To empirically evaluate CATS, we constructed a real-world O/E hybrid testbed comprising a 32-NPU cluster across four Huawei 910B compute nodes as shown in Fig. 2. The spine tier integrates a conventional 400 GbE EPS in parallel with a Polaris 7000-series 32x32 OCS. The CATS mechanism operates on a controller server, pre-configuring the topology prior to training.

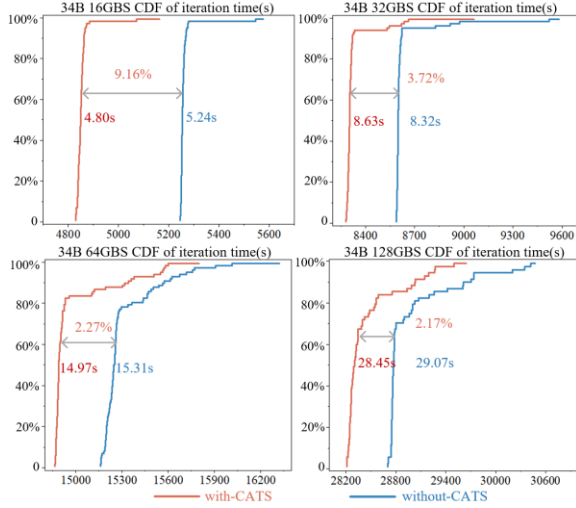


Fig. 3: CDF of iteration time in various GBS in bandwidth-constraints regime.

We evaluate a 34B parameter model across varying GBS to induce different constraints. As illustrated by the Cumulative Distribution Functions (CDF) of iteration time (T_{iter}) in Fig. 3, the CATS configuration exhibits a pronounced leftward shift, confirming strict reductions in average. In the bandwidth-constraint regime ($GBS=16$), massive DP synchronizations suffer incast on the EPS. By offloading this traffic to the OCS, CATS accelerates T_{iter} by 9.16% (5.24s to 4.80s).

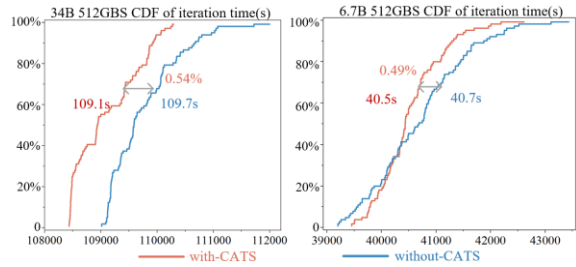


Fig. 4: CDF of iteration time in various GBS in latency-constraints regime.

Conversely, in the latency-constraint regime in Fig. 4 ($GBS=512$) where PP inter-stage jitter dominates, routing PP traffic over the zero-jitter OCS reduces T_{iter} by 0.54% (109.7s to 109.1s). Including a smaller 6.7B model, CATS consistently accelerates training by 0.49% (40.7s to 40.5s). This scenario opposes the volume-centric mechanism firmly where they will allocate optical paths for DP traffic.

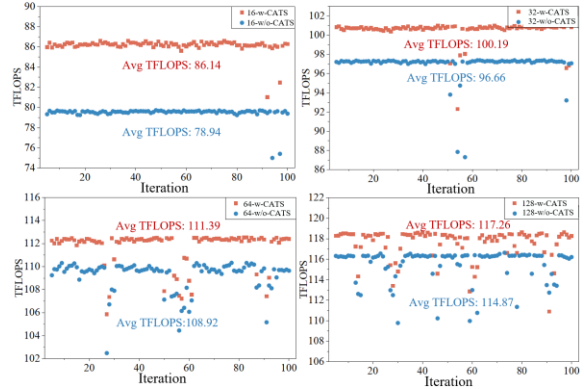


Fig. 5: Scatter distributions of iteration-level TFLOPS in bandwidth-constraints regime.

Translating network delay into computational throughput (TFLOPS per GPU) further demonstrates profound hardware utilization gains. In bandwidth-constraints regime ($GBS=16$) in Fig. 5, CATS bypasses EPS congestion and ascends average TFLOPS by 9.12% (78.94 to 86.14), exhibiting a denser scatter distribution. This eliminates extreme low-utilized GPU that previously degraded overall throughput, signifying a more uniform and exhaustive release of GPU capacity.

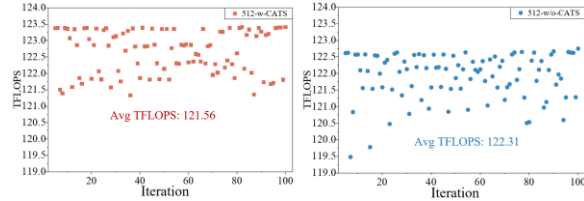


Fig. 6: Scatter distributions of iteration-level TFLOPS in latency-constraint regime.

In latency-constraint regime ($GBS=512$) in Fig. 6, the constraint inverts; by routing PP flows over zero-jitter optical paths, CATS circumvents queuing variance, improving TFLOPS by 0.62% (121.56 to 122.31). This achieves a tighter scatter distribution, guaranteeing full GPU compute utilization compared to the volume-centric mechanism. Ultimately, aligning the optical topology with the dominant constraint successfully translates into sustained computational throughput across varied workloads.

Conclusions

In this paper, we introduce CATS, a mechanism that identifies prevailing network constraints in LLM and schedules the topology to allocate the contention-free and zero-jitter optical paths specifically for constraint-inducing traffic—either bandwidth-sensitive DP or latency-sensitive PP. CATS fundamentally bypasses the blindly scheduling limitations of volume-centric mechanism. Real-world evaluations confirm this CATS accelerates training iterations by 0.49%–9.16% and boosts TFLOPS by 0.61%–9.12%, effectively improving the training efficiency in O/E hybrid switching networks.

Acknowledgements

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